

Token-Level Uncertainty as a Predictor of Hallucination in Large Language Models

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ABSTRACT

Large Language Models (LLMs) have demonstrated exceptional capabilities across diverse natural language processing tasks; however, they remain prone to generating factually incorrect or arbitrary information, a phenomenon known as hallucination. Hallucinations pose significant barriers to deploying LLMs in high-stakes domains. Uncertainty Quantification (UQ) has emerged as a promising approach for assessing the reliability of generated text without relying on external databases. Specifically, token-level uncertainty and semantic entropy offer internal signals that can reliably predict when an LLM is hallucinating. This paper explores the theoretical foundations and current methodologies for utilizing token-level uncertainty, such as Claim Conditioned Probability (CCP) and Calibrated Entropy Scores (CES), to detect hallucinations. It evaluates the shift toward efficient detection frameworks—including Semantic Entropy Probes (SEPs) and pre-trained UQ heads—and discusses the ongoing research challenges, limitations, and future directions in creating trustworthy generative models.

Keywords: Large Language Models, Hallucination Detection, Uncertainty Quantification, Token-Level Uncertainty, Semantic Entropy.

1. INTRODUCTION

The rapid integration of Large Language Models (LLMs) into real-world applications highlights both their fluency in text generation and their critical flaws. Despite their vast utility, these models have a tendency to hallucinate, producing outputs that sound plausible but are factually erroneous or entirely unfaithful to source material (Kossen et al., 2024). In domains such as medicine, law, and journalism, even occasional factual inaccuracies can be dangerous, particularly when obscured within outputs that are largely factually correct (Fadeeva et al., 2024).

Because LLMs often express high confidence even when hallucinating, identifying unreliable generations is an ongoing challenge. Traditional fact-checking pipelines rely on external knowledge retrieval, which can introduce severe computational latency and increased cost. Consequently, the research focus has shifted toward Uncertainty Quantification (UQ), which leverages the internal probabilistic states of an LLM to assess trustworthiness.

This paper provides a comprehensive analysis of token-level uncertainty as a predictor for hallucination in LLMs. It reviews how raw token probabilities, entropy distributions, and semantic-level aggregations serve as intrinsic fingerprints of hallucination. We analyze recent technical developments—from semantic entropy probes to pre-trained UQ heads—and examine the gaps that must be bridged to deploy these detection pipelines at scale.

2. BACKGROUND AND LITERATURE REVIEW

2.1 The Hallucination Problem in LLMs

A hallucination in generative models occurs when the model outputs nonsensical or unfaithful content, often resulting from a lack of epistemic alignment. The fundamental problem lies in the next-token prediction architecture. During generation, the LLM constructs an answer token by token by sampling from a learned probability distribution. When the model reaches a state where it lacks knowledge about a factual claim, it may fabricate a superficially plausible continuation. Because the generation remains syntactically robust, users

cannot distinguish the hallucinated token sequence from a faithful one based purely on surface-level reading.

2.2 Uncertainty Quantification (UQ)

Uncertainty Quantification aims to assign a confidence score to model predictions. In LLMs, uncertainty can be estimated by looking at the probability distribution over the vocabulary at each generation step. When an LLM genuinely "knows" an answer, the probability mass is generally highly concentrated on the correct sequence of tokens. When it does not, the probability mass is dispersed, resulting in higher entropy (Kossen et al., 2024).

However, measuring uncertainty in free-form natural language generation presents a distinct challenge. Token-level probabilities can be misleading because they conflate two different types of uncertainty:

- **Semantic Uncertainty:** The model's lack of confidence in the underlying factual claim or meaning.
- **Lexical/Syntactic Uncertainty:** The model's uncertainty over which specific phrasing or synonyms to use to express the answer.

If an LLM has multiple valid ways to state a correct answer, the probability of any single token might be low, generating high entropy that could be falsely flagged as a hallucination. Conversely, if a model is highly confident in an incorrect semantic claim, its token-level entropy might be deceptively low.

2.3 Entropy Distributions as Fingerprints

Recent theoretical frameworks argue that examining the entire distribution of token-level entropies throughout a generation, rather than relying solely on scalar summaries like perplexity or length-normalized entropy, provides a stronger signal for hallucination. Villani (n.d.) demonstrated that the distribution of token entropies acts as a fingerprint for hallucinated text. Specifically, when an LLM hallucinated, the underlying distribution from which the token entropies were sampled diverged significantly from the distribution observed during faithful generations. This divergence represents a measurable distributional shift and anomalies in tail behavior that provide independent signals for detection.

3. MAIN THEMATIC AND TECHNICAL DISCUSSION

3.1 Token-Level Uncertainty Quantification

Estimating uncertainty directly from the token probabilities output by the model allows for "white-box" or "grey-box" hallucination detection. Fadeeva et al. (2024) proposed a fact-checking pipeline based entirely on token-level uncertainty quantification. Their approach leverages information encapsulated in the output layers of a neural network to fact-check atomic claims within the LLM's output.

To isolate semantic uncertainty from lexical ambiguity, researchers have had to develop specialized metrics. Fadeeva et al. (2024) introduced the Claim Conditioned Probability (CCP), a novel quantification method designed to strip away the uncertainty regarding the sequence's surface form or the exact nature of the claim to be generated next. CCP is explicitly designed to measure only the uncertainty of a particular *claim value* expressed by the model. Evaluated across seven LLMs and four languages on a biography generation task, CCP demonstrated strong improvements over baseline confidence metrics, proving competitive even with fact-checking tools that leverage external knowledge bases (Fadeeva et al., 2024).

3.2 Bridging Token Uncertainty with Semantic Meaning

While raw token entropy requires careful conditioning (such as CCP) to be useful, an alternative approach is to sample multiple full-sequence generations from the LLM and analyze their semantic equivalence. This is known as Semantic Entropy (SE). If multiple generated responses to the same prompt share the same meaning (e.g., "Paris" vs. "The capital of France is Paris"), the model is semantically certain. If the responses diverge in meaning (e.g., "Paris", "Rome", "Berlin"), the model is hallucinating (Kossen et al., 2024).

While SE is highly effective at detecting hallucinations, it requires sampling 5 to 10 distinct completions for every user query. This 5-to-10-fold increase in computational cost severely hinders the practical adoption of SE in production environments (Kossen et al., 2024).

4. CURRENT APPROACHES AND DEVELOPMENTS

To combat the massive computational overhead associated with multi-sample hallucination detection, recent literature has pivoted toward lightweight, single-pass algorithms that estimate semantic uncertainty directly from internal model states.

Key Methodological Paradigms

CCP: Isolates claim uncertainty from surface lexical variations.

SEP: Approximates semantic entropy from single-pass hidden states.

CES: Applies hypothesis testing to token entropy distributions.

UQ Heads: Pre-trained auxiliary transformers mapping attention maps and logits to hallucination risks.

4.1 Semantic Entropy Probes (SEPs)

Kossen et al. (2024) introduced Semantic Entropy Probes (SEPs), which are designed to directly approximate semantic entropy from the hidden states of an LLM during a single generation pass. SEPs are linear probes trained to capture the uncertainty in semantic space. Because they rely only on the hidden states of a single model generation, they effectively reduce the computational overhead of semantic uncertainty quantification to near zero.

Empirical results demonstrated that SEPs maintain high performance in hallucination detection and generalize better to out-of-distribution data than traditional probing methods that directly attempt to predict model accuracy (Kossen et al., 2024). Crucially, the success of SEPs suggests that the hidden states of an LLM inherently encode the model's semantic uncertainty across various token positions and network layers.

4.2 Pre-trained Uncertainty Quantification Heads

Advancing beyond simple linear probes, Shelmanov et al. (2025) proposed the use of pre-trained Uncertainty Quantification heads—supervised auxiliary modules attached to LLMs. These UQ heads leverage the transformer architecture and extract highly informative features from both the attention maps and the logits of

the host LLM. Because these modules are explicitly trained to capture uncertainty, they significantly outperform standard unsupervised UQ methods in claim-level hallucination detection. Experimental data indicated that these pre-trained heads are highly robust, achieving state-of-the-art performance across both in-domain and out-of-domain prompts, and uniquely demonstrating strong generalization to languages they were not initially trained on (Shelmanov et al., 2025).

4.3 Calibrated Entropy Scores (CES)

Another prominent single-pass development is the formalization of hallucination detection as a statistical hypothesis test. Villani (n.d.) introduced the Calibrated Entropy Score (CES), an algorithm that requires only a single forward pass and basic black-box access to the LLM's token logits. CES combines the mean signal and the maximum signal of the generated entropy, evaluated against a calibrated reference cumulative distribution function (CDF) derived from non-hallucinated data. By pooling varying-length entropy sequences, CES flags deviations from the reference distribution. The method achieves performance statistically indistinguishable from expensive multi-sample methods and provides formal mathematical guarantees regarding false positive and false negative error rates (Villani, n.d.).

5. RESEARCH CHALLENGES AND LIMITATIONS

Despite the rapid evolution of UQ methods, several persistent challenges limit the absolute reliability of token-level uncertainty as a hallucination predictor:

- 1. Miscalibration in Foundational Models:** LLMs are prone to being overconfident in incorrect outputs or underconfident in correct ones. If the underlying token logits are systematically miscalibrated, methods that rely solely on these raw values—even advanced statistical methods like CES—may inherit those blind spots.
- 2. Computational Overhead vs. Accuracy:** While methods like Semantic Entropy provide highly robust hallucination detection by analyzing the diversity of multiple generated samples, their computational cost remains prohibitive for real-time systems (Kossen et al., 2024). Single-pass approximations (SEPs, UQ heads) bridge this gap

but rely heavily on supervised training data for the probes, which risks out-of-distribution degradation.

3. **Layer-Specific Discrepancies:** The internal representations of confidence change dramatically across the layers of an LLM. While probing hidden states has shown promise, identifying the precise layer and token position where epistemic uncertainty is most highly localized remains a heuristic endeavor (Kossen et al., 2024).
4. **Complex and Multi-Step Reasoning:** In tasks requiring extended logical reasoning (e.g., mathematical proofs or coding), a hallucination at an early token step can compound. Token-level uncertainty models must accurately aggregate atomic claim uncertainties over long sequences without being overwhelmed by noise.

6. RESEARCH GAPS AND FUTURE DIRECTIONS

The intersection of uncertainty quantification and generative AI presents a vast frontier for future work. Several gaps in the current literature must be addressed to construct inherently trustworthy models:

- **Unsupervised Calibration:** Current hypothesis-testing frameworks like CES require access to non-hallucinated reference data to build baseline distributions (Villani, n.d.). Future research should explore fully unsupervised methods for dynamic calibration during inference, removing the need for curated reference datasets.
- **Architectural Integration:** Rather than treating UQ heads as post-hoc auxiliary modules (Shelmanov et al., 2025), future foundation models might integrate uncertainty estimation directly into the pre-training objective. This could allow models to natively abstain from answering when their internal semantic entropy exceeds a calibrated threshold.

- **Granularity of Detection:** Most current methods operate on atomic claims or full sequences. There is a critical gap in detecting subtle, sub-claim hallucinations—such as implied biases or logical misattributions—that may not trigger a massive shift in overall token entropy.
- **Cross-Modal Generalization:** The principles of token-level uncertainty must be generalized beyond text to multimodal LLMs, establishing whether visual or audio tokens exhibit similar entropy distributions during hallucinated generations.

7. CONCLUSION

Hallucinations in Large Language Models present a significant hurdle to their safe deployment in complex and high-stakes environments. This paper has explored how Token-Level Uncertainty and Semantic Entropy serve as highly predictive intrinsic signals for detecting these fabrications. By analyzing the raw probability distributions of token generation, researchers can separate lexical ambiguity from genuine epistemic uncertainty.

Recent advancements highlight a decisive shift from computationally expensive multi-sample techniques toward lightweight, single-pass solutions. Innovations such as Claim Conditioned Probability (CCP), Semantic Entropy Probes (SEPs), the Calibrated Entropy Score (CES), and pre-trained UQ heads provide robust, mathematically grounded methods for real-time hallucination detection. While challenges such as model miscalibration and layer-specific heuristic tuning persist, the progression of UQ methodologies offers a clear path toward developing LLMs capable of introspective fact-checking and automated uncertainty mitigation.

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